

Five-layer eval engineering system

Trace -> judgment -> validation -> improvement -> operation

Operating layer

Monitoring | platform evals | cost controls | ship gates

Improvement layer

Datasets | experiments | prompt/model/tool changes

Validation layer

Golden data | human labels | meta-eval | bias checks

Judgment layer

Code eval | LLM-as-a-judge | human evaluation

Evidence layer

Instrumentation | trace/span | attributes | cost/latency

failures become
regression coverage

Phoenix is the workshop bench; Arize AX is the enterprise operating surface